

ACTL3142 Tutorial

Week 1: Introduction to Statistical Learning

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Introduction

Hi! My name is **Nick** and I'm excited to be tutoring you this term.

A little about me:

- ▶ Fourth-year Actuarial Studies and Computer Science student
- ▶ Interested in poker and basketball
- ▶ I took this course in T2 2024

Advice

This course is far too rich to learn exclusively within tutorials. Please keep up with lectures and ideally read the website and textbook. The exam is typed (Inspira), so focus on **intuition and application** rather than memorising formulas.

Icebreakers

Please go around and introduce yourself:

1. What's your name?
2. Year and degree
3. Favourite course you've taken
4. An emoji to describe your day so far

What is Statistical Learning?

A set of tools for **understanding data** and **predicting outcomes**.

We assume the data follows a model of the form:

$$Y = f(\mathbf{X}) + \varepsilon$$

- ▶ f : an unknown function capturing the **systematic** relationship between inputs and output
- ▶ $\mathbf{X} = (X_1, \dots, X_p)$: the predictors (features, independent variables)
- ▶ Y : the response (output, dependent variable)
- ▶ ε : the irreducible error, everything $f(\mathbf{X})$ cannot explain (random chance, measurement error, missing variables)

Our goal: estimate f . The resulting $\hat{f}$ is our **model**.

Prediction vs Inference

Prediction

- ▶ Goal: accurately estimate Y given X
- ▶ The form of f is not the focus; we just want accurate outputs
- ▶ Models can be very complex (a black box is fine)

Inference

- ▶ Goal: understand *how* X affects Y
- ▶ Which predictors matter? What is their individual effect?
- ▶ Models tend to be simpler and more interpretable

Questions to always ask

What predictors should we include? Do we want to predict or explain? What is signal and what is noise?

Key Distinctions

- ▶ **Supervised** (Y observed) vs **Unsupervised** (no Y ; find structure)
- ▶ **Regression** (Y quantitative) vs **Classification** (Y categorical)
- ▶ **Interpretability** vs **Flexibility**: easy to understand f , or able to capture complex relationships (usually a tradeoff)

Two approaches to estimating f :

Parametric

- ▶ Assume a form for f (e.g. linear)
- ▶ Estimate a few parameters
- ▶ Works with limited data
- ▶ Risk: misses signal (**high bias**)

Non-parametric

- ▶ No assumption about f 's shape
- ▶ Many effective "parameters"
- ▶ Needs lots of data
- ▶ Risk: fits noise (**high variance**)

These distinctions frame the whole course: most of what we study

CQ1: Flexible vs Inflexible Methods (ISLR2 Q2.1)

1. ★ (ISLR2, Q2.1) For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.
 - a. The sample size n is extremely large, and the number of predictors p is small.
 - b. The number of predictors p is extremely large, and the number of observations n is small.
 - c. The relationship between the predictors and response is highly non-linear.
 - d. The variance of the error terms, i.e., $\sigma^2 = \mathbb{V}(\epsilon)$, is extremely high.

CQ2: Classification or Regression? (ISLR2 Q2.2)

2. (ISLR2, Q2.2) Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide n and p .
- a. We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.
 - b. We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a **success** or **failure**, price charged for the product, marketing budget, competition price, and ten other variables.
 - c. We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Loss Functions and the Bias-Variance Tradeoff

Our model predicts $\hat{f}(x_0)$. We measure how far this is from the truth using a **loss function** (here, squared error).

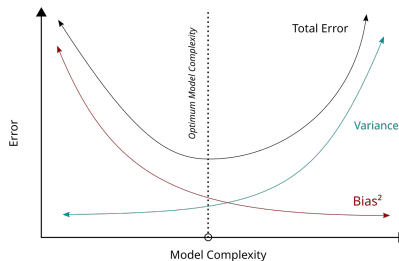
The expected test MSE decomposes as:

$$\underbrace{E[(y_0 - \hat{f}(x_0))^2]}_{\text{Expected test MSE}} = \underbrace{[\text{Bias}(\hat{f}(x_0))]^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{Variance}} + \underbrace{\text{Var}(\varepsilon)}_{\text{Irreducible}}$$

- ▶ **Bias**: systematic error from simplifying assumptions
- ▶ **Variance**: how much $\hat{f}$ changes across different training sets
- ▶ $\text{Var}(\varepsilon)$: irreducible error, a floor we cannot beat

All three terms are non-negative, so the irreducible error sets a hard lower bound on test MSE no matter how good the model is.

Visualising the Tradeoff



- ▶ More flexible $\Rightarrow$ less bias, more variance
- ▶ Less flexible $\Rightarrow$ more bias, less variance
- ▶ The best model sits at the **minimum of total error**, not at either extreme

CQ3: Bias-Variance Decomposition (ISLR2 Q2.3)

3. ★ (ISLR2, Q2.3) We now revisit the bias-variance decomposition.
- a. Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x -axis should represent the amount of flexibility in the method, and the y -axis should represent the values for each curve. There should be five curves. Make sure to label each one.
 - b. Explain why each of the five curves has the shape displayed in part (a).

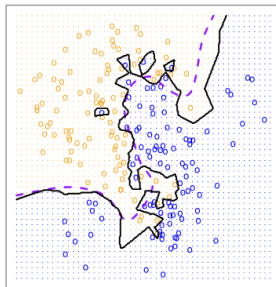
CQ4: Advantages and Disadvantages of Flexibility (ISLR2 Q2.5)

4. ★ (ISLR2, Q2.5) What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

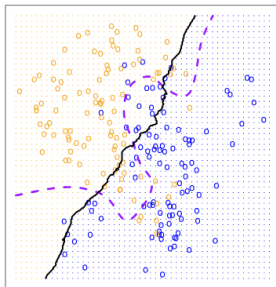
Bias-Variance in Action: KNN

KNN classifier: to classify a new point, find its K nearest neighbours and let them vote. The majority class wins.

KNN: $K=1$



KNN: $K=100$



$K = 1$ (very flexible)
Low bias, high variance $\rightarrow$
overfitting

$K = 100$ (inflexible)
High bias, low variance $\rightarrow$
underfitting

K controls flexibility directly: small K tracks every wiggle in the data, large K smooths everything out. Some K in between is best.

Simple Linear Regression: A Parametric Example

A **parametric** approach. We assume a form for f :

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Interpreting the coefficients:

- ▶ β_0 (intercept): the predicted value of y when $x = 0$
- ▶ β_1 (slope): the average change in y per one-unit increase in x

Where it sits: inflexible (we impose linearity) but easy to interpret, a clear example of the flexibility-interpretability tradeoff.

Estimating the coefficients: pick the β 's that **minimise the loss**. Using squared-error loss (MSE) gives ordinary least squares.

This is exactly where we pick up next week, in much more detail.

Applied Questions: Exploratory Data Analysis

The applied exercises are about getting comfortable with data in R before any modelling. Pull them up on your side.

Key pointers:

- ▶ `summary()`: per-variable summary statistics and counts of NAs
- ▶ `pairs()`: scatterplot matrix to eyeball relationships between variables
- ▶ `hist()`: check the distribution and skew of a single variable
- ▶ `plot()`: scatterplots, and diagnostic plots once a model is fitted

Look at the data before fitting anything. Outliers, skew, and missing values all change how you would model, and they are far easier to catch now than after.

Thanks for coming! See you next week.

